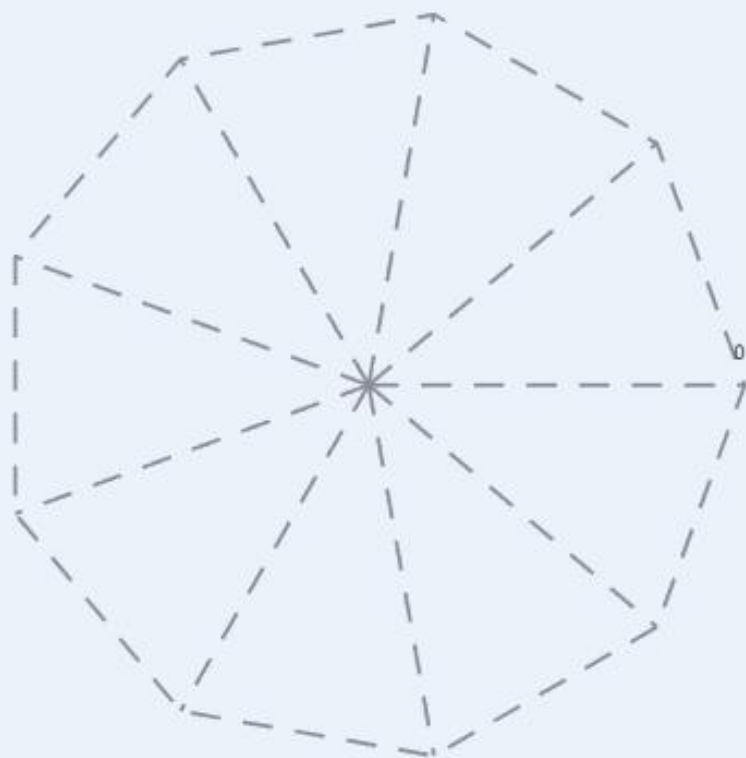




Choose at most three topics that you find most difficult (multiple vote on a single choice is allowed; max 3 votes)



- A 0% Find Cartesian from Parametric Equations
- B 0% Find Tangent Lines to Curve
- C 0% Implicitly defined parametrizations
- D 0% Area enclosed by curve
- E 0% Length of Curves
- F 0% Graph the set of polar coordinates points
- G 0% Polar to Cartesian Equations
- H 0% Cartesian to Polar Equations
- I 0% Nothing is difficult



Ex 10.2 Q17

$$x + 2x^{3/2} = t^2 + t \Rightarrow \frac{dx}{dt}$$

$$y\sqrt{t+1} + 2t\sqrt{y} = 4 \Rightarrow \frac{dy}{dt}$$

$$t = 0$$

Aim: $\boxed{\frac{dy}{dx} \big|_{t=0}}$

$$\rightarrow \frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

If $\frac{dy}{dx}$ contains x or y , set $t=0$
to find values of x or y

$$\frac{dx}{dt} : \quad \underline{x} + 2 \underline{x^{3/2}} = t^2 + t$$

functions of t

$$\begin{aligned} \frac{d}{dt} x &= \frac{dx}{dt}, & \frac{d}{dt} 2x^{3/2} &= 2 \frac{d}{dt} \underbrace{x^{3/2}}_{[\underline{x(t)}]^{3/2}} \\ & & &= 2 \cdot \frac{3}{2} x^{1/2} \cdot \frac{dx}{dt} \end{aligned}$$

$$x + 2x^{3/2} = t^2 + t$$

$$\Rightarrow \underbrace{\frac{dx}{dt}} + 3x^{1/2} \underbrace{\frac{dx}{dt}} = 2t + 1$$

$$\Rightarrow \frac{dx}{dt} (1 + 3x^{1/2}) = 2t + 1$$

$$\Rightarrow \underline{\underline{\frac{dx}{dt}}} = \frac{2t + 1}{1 + 3\underline{x^{1/2}}}$$

$$\frac{dy}{dt} : \underline{y} \sqrt{t+1} + 2t \underline{\sqrt{y}} = 4$$

Product Rule

$$\frac{d}{dt} (\underline{y} \sqrt{t+1}) = \frac{dy}{dt} \sqrt{t+1} + y \cdot \frac{1}{2\sqrt{t+1}}$$

$$\frac{d}{dt} (\underline{2t} \sqrt{y}) = 2 \cdot \sqrt{y} + \cancel{2t} \cdot \frac{1}{\cancel{2}\sqrt{y}} \cdot \frac{dy}{dt}$$

$$\underline{\frac{d}{dt} \sqrt{y}} = \frac{1}{2\sqrt{y}} \cdot \frac{dy}{dt} \rightarrow \text{Chain Rule}$$

$$\frac{dy}{dt} : \underline{y \sqrt{t+1}} + 2t \underline{\sqrt{y}} = \underline{4}$$

Product Rule

$$\frac{d}{dt} (\underline{y \sqrt{t+1}}) = \frac{dy}{dt} \sqrt{t+1} + y \cdot \frac{1}{2\sqrt{t+1}}$$

$$\frac{d}{dt} (\underline{2t \sqrt{y}}) = 2 \cdot \sqrt{y} + \cancel{2t} \cdot \frac{1}{\cancel{2}\sqrt{y}} \cdot \frac{dy}{dt}$$

Diff. w.r.t t:

$$\frac{dy}{dt} \underline{\sqrt{t+1}} + y \cdot \underline{\frac{1}{2\sqrt{t+1}}} + 2 \cdot \sqrt{y} + \underline{\frac{t}{\sqrt{y}} \frac{dy}{dt}} = 0$$

$$\frac{dy}{dt} = \frac{-\frac{y}{2\sqrt{t+1}} - 2\sqrt{y}}{\sqrt{t+1} + \frac{t}{\sqrt{y}}}$$

When $t = 0$, $x + 2x^{3/2} = \underline{t^3} + \underline{t}$

$$\therefore x + 2x^{3/2} = 0$$

$$x(1 + 2\sqrt{x}) = 0$$

$$\underbrace{1 + 2\sqrt{x}}_{\geq 1} \Rightarrow x = 0$$

$$\text{When } t=0, \quad y\sqrt{t+1} + 2t\sqrt{y} = 4$$

$$y\sqrt{1} = 4$$

$$\Rightarrow \underline{y = 4}$$

$$\left. \begin{array}{l} x = 0 \\ y = 4 \\ t = 0 \end{array} \right\} \frac{dy}{dx} \Big|_{t=0} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \Big|_{t=0}$$

$$= \frac{\boxed{}}{\boxed{}}$$

sub. values of
 x, y, t

Ex. 10.2 Q12 $\frac{dx}{dt}$ $\frac{dy}{dt}$

$$x = \cos t, y = 1 + \sin t, t = \pi/2$$

Aim: Line tangent to this curve at
point defined by $t = \pi/2$

$$y - \underline{y_0} = \underline{m}(x - \underline{x_0})$$

$$\left. \frac{dy}{dx} \right|_{t=\pi/2} = \frac{dy/dt}{dx/dt} \Big|_{t=\pi/2}$$

$$\left. \begin{aligned} x_0 &= x(\pi/2) \\ y_0 &= y(\pi/2) \end{aligned} \right\} \begin{array}{l} \text{values} \\ \text{obtained} \\ \text{by setting} \\ t = \pi/2 \end{array}$$

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right)$$

$$= \frac{dy'}{dx}$$

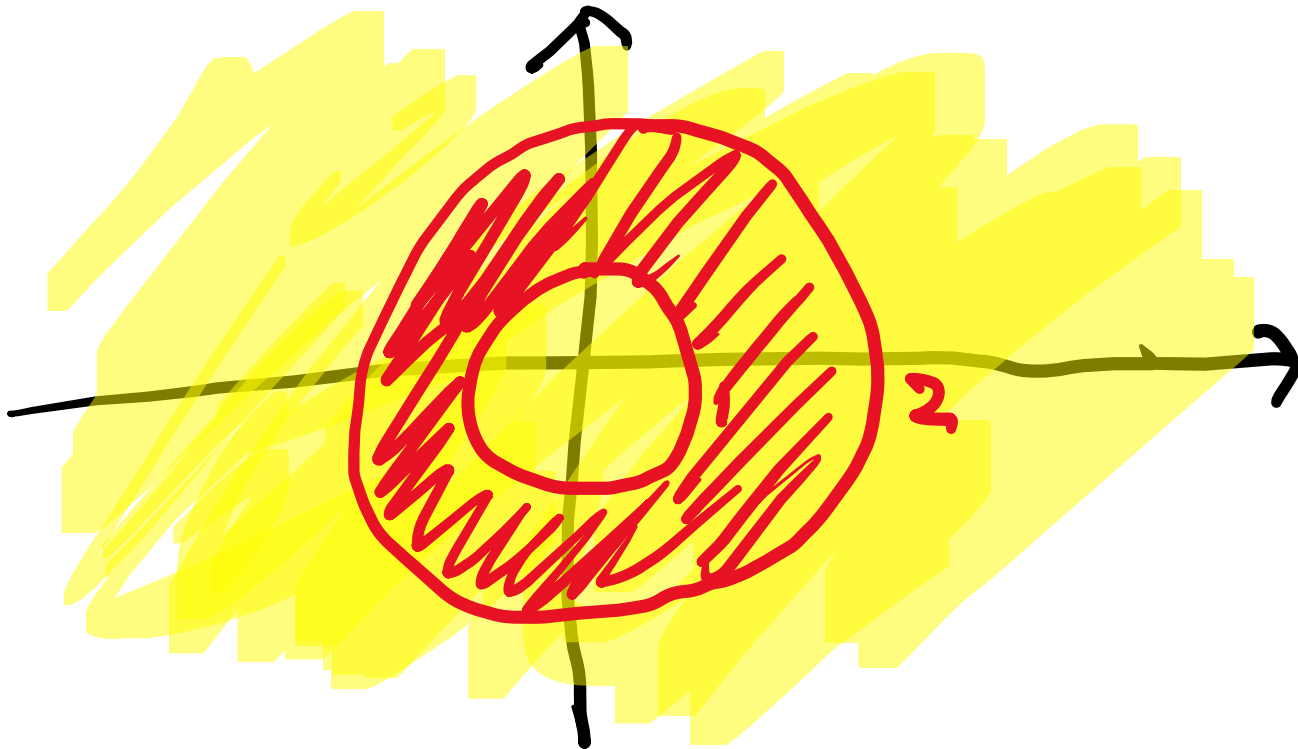
$$= \frac{dy' / dt}{dx / dt}$$

obtained by diff. $\frac{dy}{dt}$

$dx / dt \rightarrow$ obtained just now

Ex. 10.3 Q14

$1 \leq r \leq 2$ (θ can be any value)



Q 15

$0 \leq \theta \leq \pi/6$

$r \geq 0$

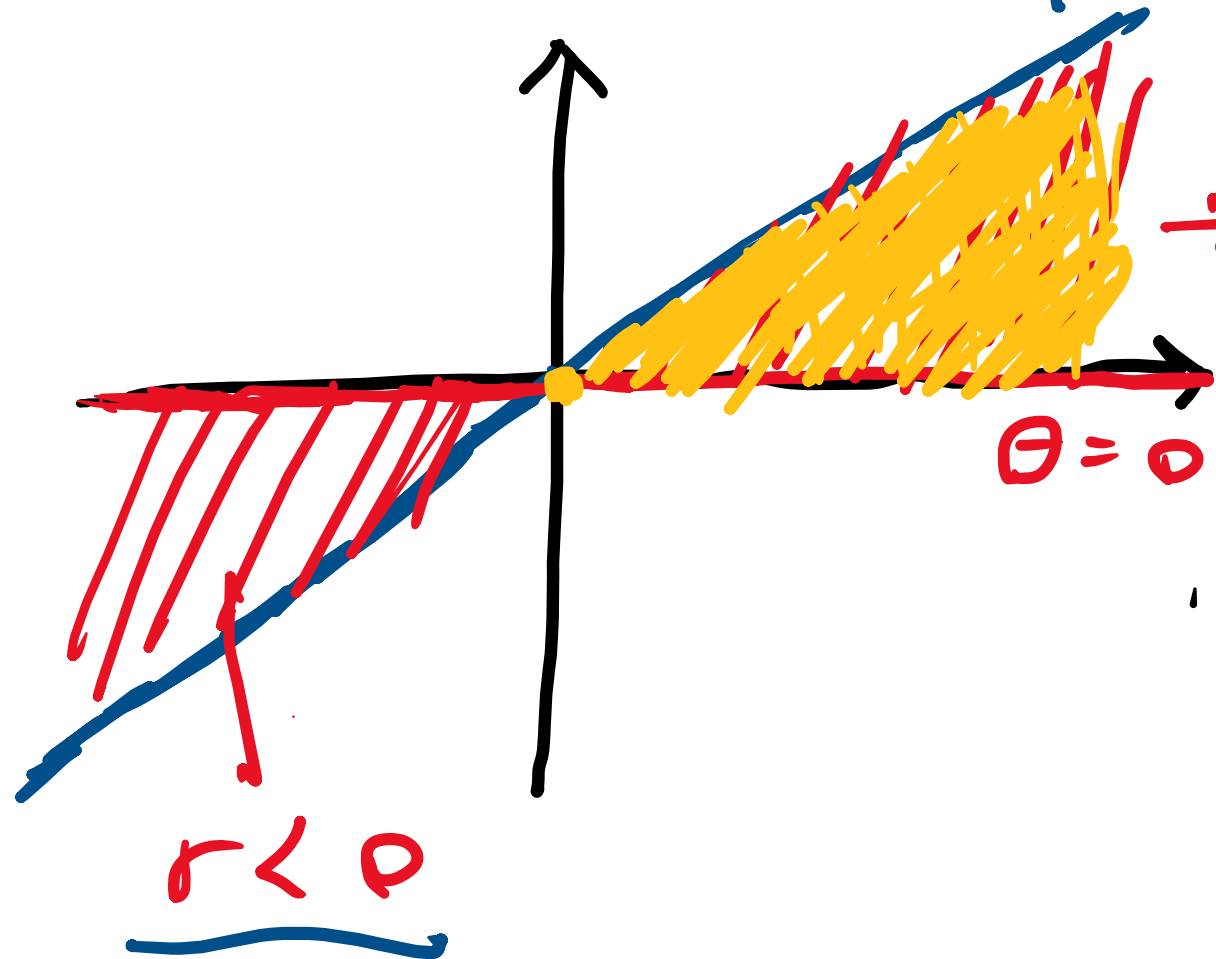
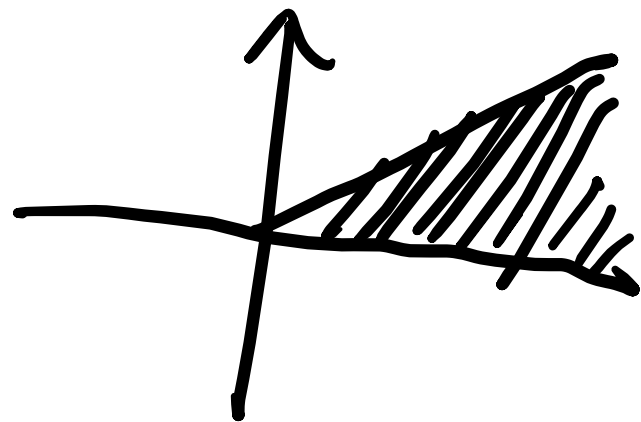
$\rightarrow \theta = \pi/6$

$\rightarrow r \geq 0$

$\theta = 0$

$\Rightarrow$

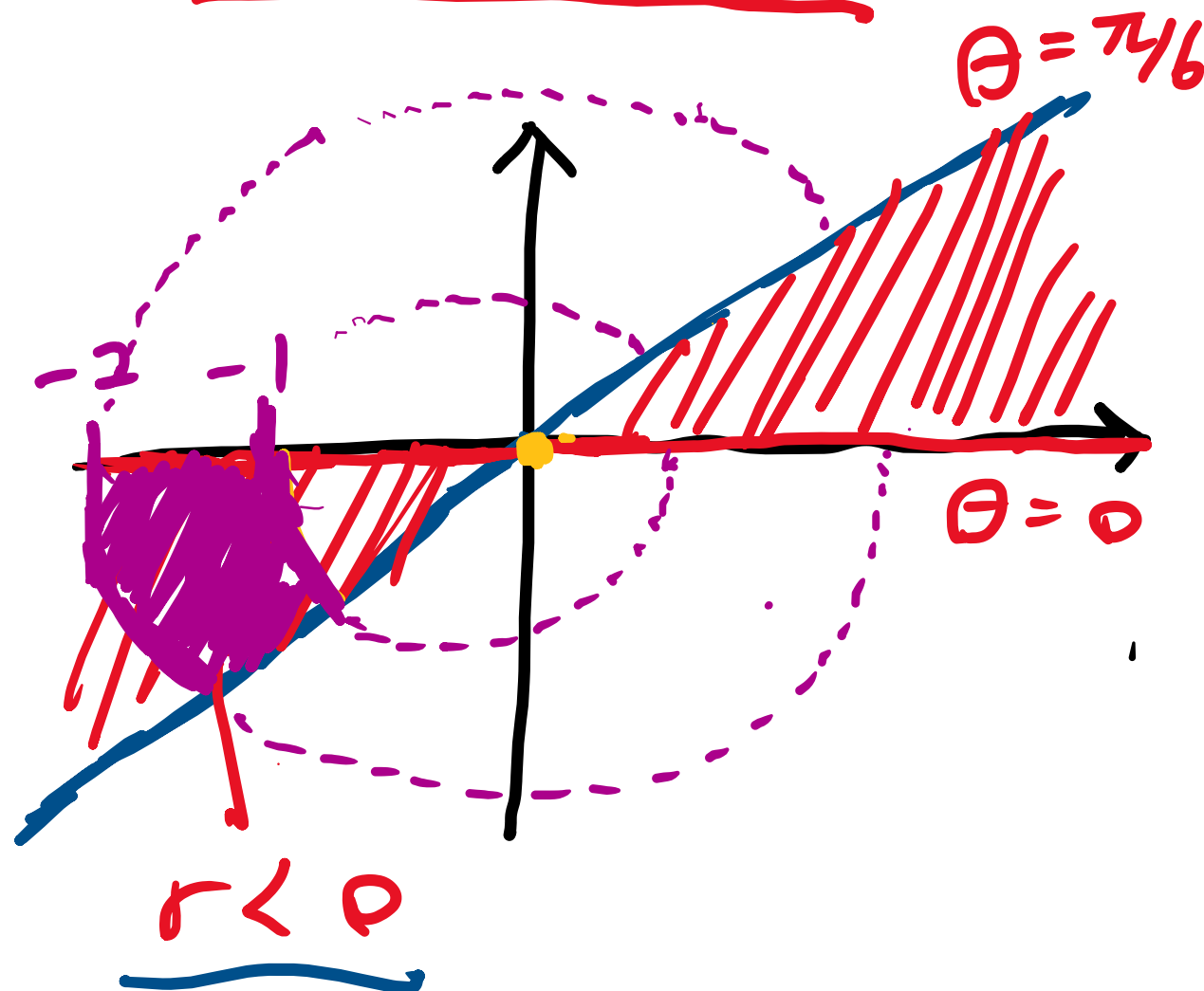
Ans :



Q 15

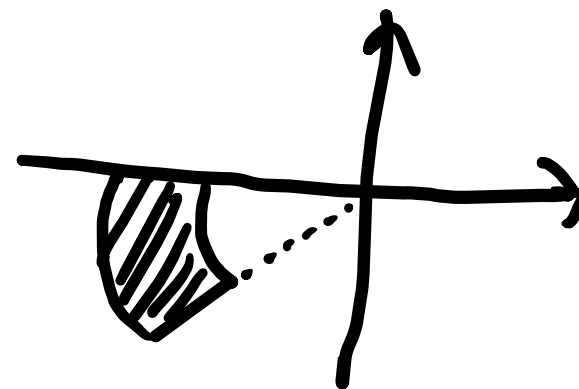
$$\underline{0 \leq \theta \leq \pi/6}$$

$$\boxed{-2 \leq r \leq -1}$$



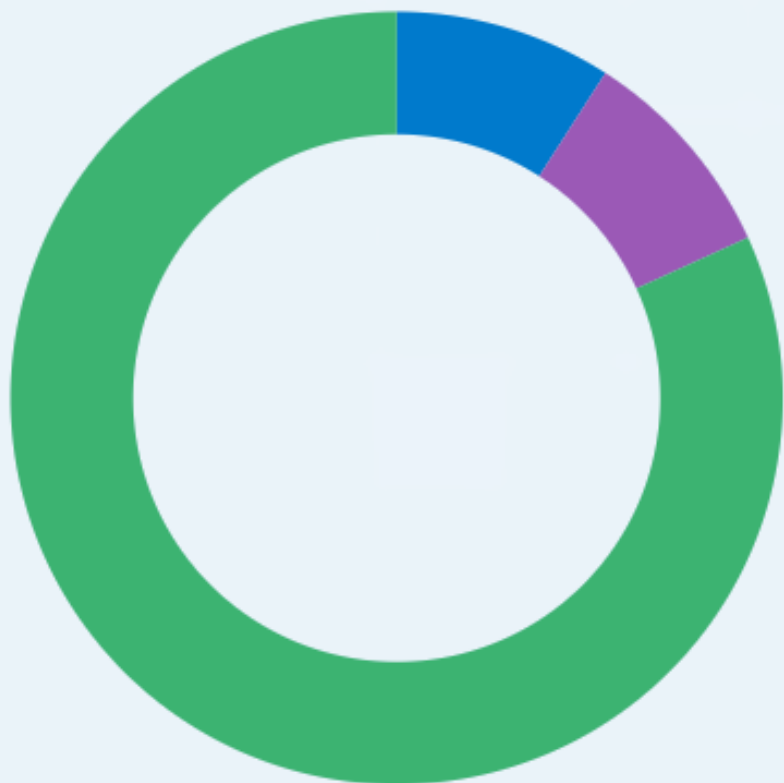
$\Rightarrow$

Ans:





Choose the correct graph for $0 \leq \theta \leq \pi/2$, $1 \leq |r| \leq 2$



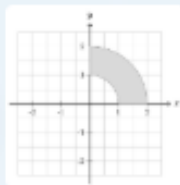
A

9%



B

9%



C

82%



D

0%

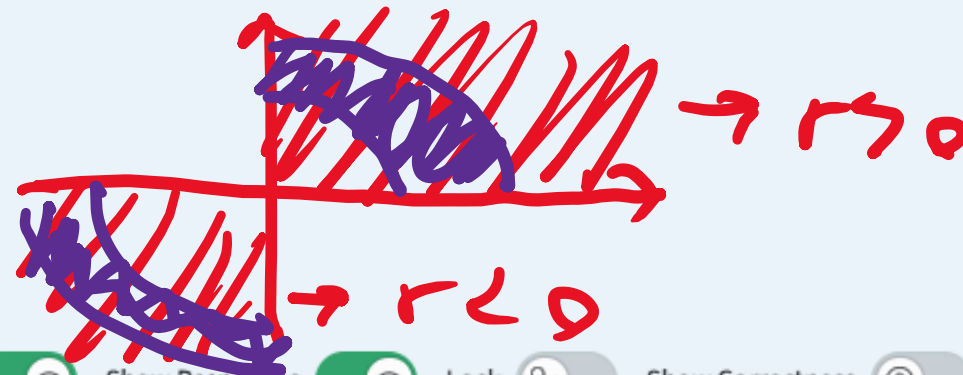


$1 \leq r \leq 2$

or

$1 \leq -r \leq 2$

$-2 \leq r \leq -1$



11 0

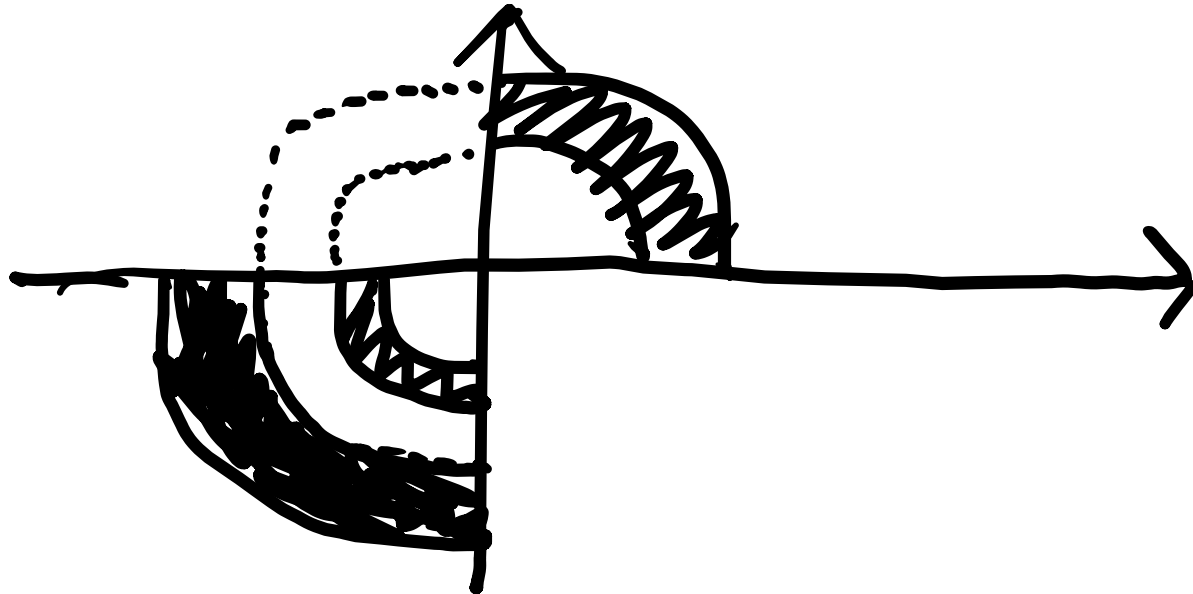


Q26 (Modified)

$$0 \leq \theta \leq \pi/2, \quad -1 \leq r \leq -\frac{1}{2}$$

$$-3 \leq r \leq -2$$

$$1 \leq r \leq 2$$



Cartesian to Polar

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$r^2 = \underbrace{x^2 + y^2}$$

Ex. 19.3 Q66

$$(x+2)^2 + (y-5)^2 = 16$$

$$x^2 + 4x + 4 + y^2 - 10y + 25 = 16$$

$$\underline{x^2 + y^2} + 4x - 10y = -13$$

$$r^2 + \underline{4r \cos \theta} - \underline{10r \sin \theta} = \underline{-13}$$

they can be zero

$$\boxed{r^2 + 4r \cos \theta - 10r \sin \theta + 13 = 0}$$